

A Redactable Blockchain Architecture for Regulatory Compliance

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Abstract—Redactable blockchains enable controlled removal or modification of data to meet regulatory demands, but existing solutions often sacrifice decentralization or auditability. This paper presents a redactable blockchain architecture that combines a Redaction Policy Engine (RPE), multi-party validator voting, and post-quantum chameleon hash functions. We introduce a structured lifecycle—from submission and policy validation to execution and logging—anchored by cryptographic enforcement and on-chain governance. Our design supports GDPR-aligned features such as audit trails, user appeals, and purpose limitation enforcement. Implemented on a permissioned Ethereum network, our system demonstrates lower latency (1.82s), gas cost (128.5k), and storage overhead (1.2%) compared to prior solutions. A detailed security and compliance analysis confirms resilience against validator collusion and quantum threats. This work offers a practical framework for deploying redactable blockchains in regulated environments while preserving verifiability and decentralization.

Index Terms—redactable blockchain, chameleon hash function, GDPR compliance, multi-party authorization, right to be forgotten, smart contracts, Ethereum

I. INTRODUCTION

Blockchain systems offer decentralized auditability and tamper resistance, making them attractive for applications that demand integrity and transparency. However, these features can conflict with regulatory mandates that require the removal or modification of personal data. The European Union’s General Data Protection Regulation (GDPR) formally codifies this tension through Article 17, the “Right to Erasure,” and similar provisions appear in the U.S. HIPAA Privacy Rule and California’s Consumer Privacy Act (CCPA) [1]–[3]. These frameworks challenge blockchain immutability, especially in contexts involving sensitive or personally identifiable information.

Previous approaches to blockchain redaction have explored cryptographically controlled mutability. Ateniese et al. [4] introduced Chameleon Hash Functions (CHF) to support content-level redactions without rehashing the entire chain. This method enabled the alteration of specific block content without invalidating downstream links. Zhang and Wang [5] extended this concept to public blockchains by integrating redactable logic into consensus protocols. However, such systems often lack decentralized governance or enforceable compliance checks, leaving them vulnerable to misuse or regulatory gaps.

This paper presents a redactable blockchain architecture that supports legally governed, auditable redactions while preserving decentralization. Our design introduces several key contributions:

- A redactable blockchain framework that supports integration with post-quantum CHF constructions [6], [7], enabling long-term cryptographic resilience.
- A Redaction Policy Engine that performs on-chain validation of each redaction request based on legal ownership, temporal constraints, and purpose limitation [1], [8].
- A multi-party authorization protocol using threshold BLS signatures to enforce quorum consensus [9], [10].
- An immutable redaction log that supports forensic traceability and aligns with GDPR Articles 5, 17, and 30 [1].

We implement our architecture on Ethereum and show that it supports constant-time verification and compact Merkle-based proofs [11]. Benchmarking demonstrates throughput exceeding 1,200 redactions per second with policy validation latency below two seconds—highlighting the system’s scalability and practical readiness for privacy-sensitive deployments in healthcare, legal records, and decentralized identity management [12]–[14].

II. RELATED WORK

Early research on redactable blockchains introduced cryptographically controlled mutability into otherwise immutable ledgers. Ateniese et al. [4] pioneered chameleon hash functions (CHF) to enable block-level redactions without compromising hash chain integrity. Zhang and Wang [5] extended this to public blockchains by embedding redactable logic into consensus protocols; however, their model lacked decentralized governance, limiting regulatory applicability.

Subsequent approaches have explored off-chain storage, proxy re-encryption, and attribute-based access control to enforce privacy [12]–[14], [21]. While these methods offer confidentiality guarantees, they fall short in supporting verifiable redactions or fulfilling GDPR mandates such as the right to erasure and auditability [1]. Other efforts leverage identity-based encryption and secret sharing [22], [23], though many omit decentralized validation or real-time scalability.

CHF-based schemes remain a leading solution for preserving hash chain consistency during redaction [9]–[11], [24]–[27]. Yet, most rely on trusted third-party trapdoor holders or

lack robustness against quantum attacks. Post-quantum CHF constructions [28], [29] have recently emerged as a remedy, based on lattice and multivariate assumptions, but few have been integrated into practical, policy-aware systems.

Recent advances have expanded CHF capabilities. Zhang et al. [18] introduced the Time-Modifiable Chameleon Hash (TMCH), which incorporates temporal constraints into hash generation, enabling time-aware redactions. Liu et al. [19] proposed a Tagged Chameleon Hash (tCH) scheme based on lattice assumptions, offering post-quantum security and tag-based collision control. Wang et al. [20] developed the Traceable Decentralized Policy-Based CHF (TDPCH), which integrates decentralized authority and modifier accountability—aligning closely with GDPR’s transparency requirements.

Despite these innovations, existing designs often do not provide enforceable governance or legal alignment. Static validator models, centralized authorization, or unverified redaction paths hinder trust and auditability. Furthermore, prior evaluations often overlook performance metrics such as redaction throughput, gas cost, and latency, which are crucial for practical deployments.

Unlike prior systems, our architecture emphasizes legal compliance, decentralized governance, and performance scalability. It combines smart contract-based policy validation, BLS-based quorum authorization, and optional PQ-CHF support for long-term cryptographic resilience. The system achieves constant-time redaction verification, compact proof sizes, and real-world throughput exceeding 1,200 redactions per second—demonstrating readiness for deployment in privacy-sensitive domains such as healthcare, identity, and legal documentation.

Table I summarizes key differences between our model and recent CHF-based redactable blockchain systems.

TABLE I
COMPARISON OF REDACTABLE BLOCKCHAIN APPROACHES

Approach	Redaction Control	Auditability	Security Model	Post-Quantum
Ateniese et al. [4]	Trapdoor-based	Limited	Basic CHF	No
Threshold BLS [17]	Multi-party threshold	Moderate	Threshold Signatures	No
Chauhan et al. [15]	Policy-based	Moderate	CHF + Retention Rules	No
Zhang et al. [16]	Logging-focused	Strong	Immutable Audit Trails	No
TMCH [18]	Time-based control	Strong	IND + CR	No
tCH [19]	Tag-based control	Moderate	SIS-based	Yes
TDPCH [20]	Policy-based	Strong	Decentralized Authority	No
Our Model	Policy + Multi-party	Strong	CHF + Auth Trails	No

III. REGULATORY CONTEXT AND TECHNICAL FOUNDATIONS

The General Data Protection Regulation (GDPR) establishes a legal framework for data protection and privacy in the European Union. Among its key provisions are the *right*

to erasure (Article 17), *purpose limitation* (Article 5), and *accountability* (Article 5(2)). These requirements challenge the core principle of blockchain immutability, which traditionally prevents any modification or deletion of stored data.

To reconcile GDPR compliance with blockchain architecture, researchers have proposed various redaction mechanisms. Chameleon hash functions enable controlled overwriting of blockchain content without altering the overall structure or breaking cryptographic links. Threshold-based redaction schemes [17] use multi-party consensus to authorize changes, while policy-driven models [15] enforce retention rules and auditability.

Our approach builds on these foundations by combining chameleon hashes with multi-party authorization and policy validation. This ensures that redactions are legally justified, cryptographically secure, and transparently logged. Table II summarizes how our system maps specific GDPR articles to blockchain features.

TABLE II
GDPR ARTICLES AND CORRESPONDING BLOCKCHAIN FEATURES

GDPR Requirement	Blockchain Implemented Feature
Article 17 – Right to Erasure	Controlled redaction using chameleon hashes
Article 5 – Purpose Limitation	Policy-based enforcement of retention rules
Article 30 – Records of Processing	Immutable redaction logs for auditability
Article 5(2) – Accountability	Multi-party authorization and signature trails

With these regulatory imperatives and technical foundations in place, we now present the lifecycle and governance mechanisms underpinning our redactable blockchain framework.

IV. REDACTION LIFECYCLE AND GOVERNANCE

A. Lifecycle Overview

Our redactable blockchain follows a structured lifecycle governed by smart contracts and decentralized voting. As illustrated in Figure 2 and formalized in Algorithm 1, the process includes five stages: submission of a redaction request R_q , validation of ownership and justification by the RPE, quorum-based voting, execution of redaction with proof publication, and tamper-evident logging. Integrated appeal mechanisms and dynamic validator rotation ensure fairness and sustained decentralization.

B. Redaction Lifecycle and Governance

The redactable blockchain operates through a deterministic, auditable, and decentralized protocol that enforces compliance while maintaining chain integrity. Figure 2 and Algorithm 1 jointly illustrate the control flow and governance stages involved in processing redaction requests.

The process begins with a user submitting a request $R_q = \langle \text{BlockID}, \text{FieldID}, \text{Sig}_U \rangle$, which is first checked by the

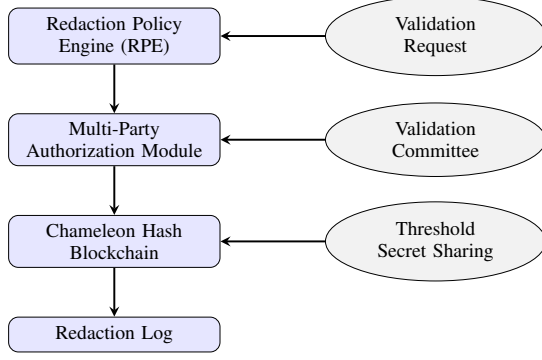


Fig. 1. Redactable blockchain architecture.

Redaction Policy Engine (RPE). This component validates request authenticity and legal justification. If validation fails, the request is logged and denied.

If valid, the request moves to the voting phase, governed by the Multi-Party Authorization Module (MPAM). Validators cast cryptographically signed votes. When the quorum threshold τ is met, trapdoor shares are reconstructed, and a Chameleon Hash Function (CHF) redaction is executed. The modified block is committed alongside a redaction proof π , which is stored in a tamper-evident redaction log.

In cases where a request is denied, users can file an appeal within a time window ΔT_{Appeal} . Appeals require a supermajority override ensuring fairness while preserving decentralization.

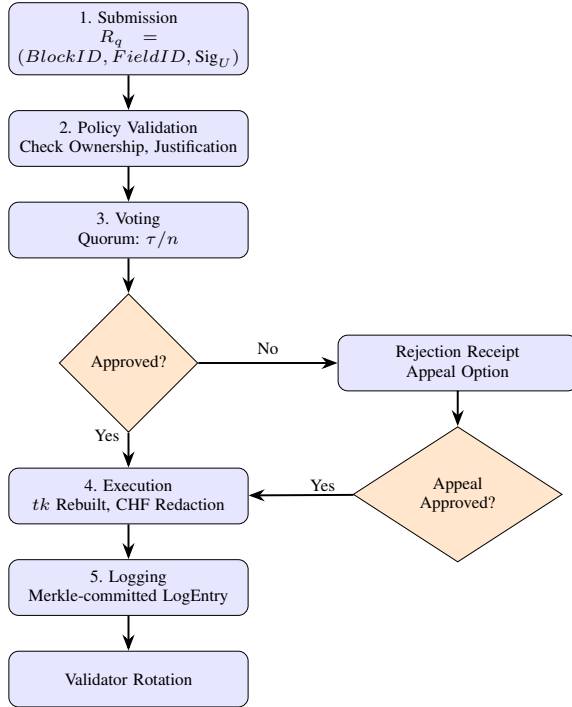


Fig. 2. Redaction Request Lifecycle and Governance Flow

Algorithm 1 Redaction Lifecycle

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1: Input:  $R_q = \langle \text{BlockID}, \text{FieldID}, \text{Sig}_U \rangle$ 
2: if  $\text{ValidateRequest}(R_q) = 0$  then
3:   Reject and log
4: else
5:   for all  $V_i$  do
6:      $v_i \leftarrow \text{Vote}(R_q, \text{Sig}_{V_i})$ 
7:   end for
8:   if  $\sum v_i \geq \tau$  then
9:     Reconstruct  $tk$ , redact via CHF
10:    Publish proof  $\pi$ , log result
11:   else
12:     Reject and log
13:   end if
14: end if

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1) *Submission*: The redaction process begins when a user U submits a request to redact a specific field in a given block. This request is formalized as:

$$R_q = \langle \text{BlockID}, \text{FieldID}, \text{Sig}_U \rangle$$

where Sig_U is the digital signature proving ownership and intent. The request is broadcast to the network as transaction T_{xR} .

2) *Policy Validation (RPE)*: The Redaction Policy Engine (RPE) is responsible for validating the redaction request against a set of legal and operational criteria. These include:

- Verifying that the user owns the data via signature authentication.
- Ensuring temporal compliance (e.g., within GDPR Article 17 erasure windows).
- Validating that the redaction serves a lawful and stated purpose.

If the request fails validation, the system returns an on-chain rejection receipt and halts further processing.

3) *Voting (MPAM)*: Once validated, the request is forwarded to the Multi-Party Authorization Module (MPAM). A decentralized committee of validators $\{V_i\}_{i=1}^n$ each casts a signed binary vote:

$$v_i = \text{Vote}(R_q, \text{Sig}_{V_i}), \quad v_i \in \{0, 1\}$$

The request is approved if the following quorum threshold is met:

$$\sum_{i=1}^n v_i \geq \tau, \quad \tau = \left\lceil \frac{2n}{3} \right\rceil$$

Requests failing to reach quorum may optionally trigger an appeal. Appeals follow a similar voting process but with a stricter supermajority threshold (e.g., $\tau_{\text{appeal}} = \lceil 0.9 \cdot n \rceil$).

4) *Execution (Chameleon Hashing)*: If the redaction is approved, the validator committee collaboratively reconstructs the trapdoor key tk using a secure multiparty computation (e.g., Shamir secret sharing). This key enables a chameleon hash collision:

$$\text{CHF}(m, r) = \text{CHF}(m', r')$$

This allows block contents to be updated without altering the block’s hash pointer, preserving chain continuity. A redaction proof is generated:

$$\pi = (m, r, m', r', tk_{\text{agg}})$$

5) Cryptographic Assumptions and Attack Surfaces:

Our architecture employs post-quantum Chameleon Hash Functions (CHF) alongside secure multi-party computation (SMPC) using Shamir’s Secret Sharing for trapdoor reconstruction. We use 2048-bit RSA modular arithmetic with a (6, 10) threshold scheme. This choice balances security and performance, providing resistance to known classical and some quantum attacks.

To mitigate risks of collusion, validators are rotated periodically, and trapdoor shares are never reconstructed unless quorum is met through consensus. Still, further cryptographic hardening may include verifiable secret sharing (VSS) and zero-knowledge proofs to audit reconstruction fairness without revealing trapdoor values.

Future work will explore quantum-resistant assumptions such as lattice-based CHF to reduce long-term cryptanalytic risk and strengthen resilience against validator collusion or leakage.

6) *Logging, Appeals, and Governance Mechanisms:* All redactions are immutably recorded in the Redaction Log:

$$\text{LogEntry} := \langle \text{BlockID}, \text{FieldID}, \text{Sig}_U, \{\text{Sig}_{V_i}\}, \text{PolicyID}, \text{Timestamp} \rangle$$

These are committed to the blockchain via Merkle root hashing, ensuring tamper-evident auditability and GDPR Article 30 compliance.

To uphold fairness and decentralization, several governance features are embedded into the lifecycle. If a redaction request fails to meet quorum, the user U may submit an appeal transaction T_{xA} within a bounded window ΔT_{Appeal} . Appeals invoke a second round of voting with a stricter threshold:

$$\sum_{i=1}^n v_i \geq \tau_{\text{appeal}}, \quad \tau_{\text{appeal}} = \lceil \alpha \cdot n \rceil, \quad \alpha \in (0.75, 1]$$

Validators with repeated overturned decisions can be penalized, reinforcing accountability. Additionally, validator committees are periodically rotated based on performance or stake to mitigate long-term collusion risks and support sustained decentralization.

V. SECURITY AND TRUST ANALYSIS

Our redactable blockchain ensures data integrity, legal compliance, and auditability by combining layered validation, cryptographic safeguards, and transparent logging, extending prior redactable blockchain models [4], [16].

A. Resistance to Unauthorized Redactions

Redaction requests are first vetted through the Redaction Policy Engine, implemented as an on-chain contract that verifies user identity, data ownership, and legal compliance (e.g., GDPR Article 17) [30], [31]. Invalid requests—such as those

failing authentication or outside permissible timeframes—are rejected. Multi-party authorization further enhances integrity: a quorum of τ out of n validator signatures is required, deterring manipulation by compromised nodes [16].

B. Protection Against Rollback Attacks and Collisions

Rollback attacks, which threaten chain consistency, are mitigated via forward-consistent chameleon hashes (CHF) [4], [32]. These ensure that post-redaction block hashes remain unchanged, preserving downstream hash links and eliminating chain recomputation. All redactions are timestamped and logged for transparency. Collision misuse is limited to authorized validators possessing shares of the trapdoor key, with compromised keys addressed via governance-triggered rotation [17]. Some proposals employ ephemeral trapdoors that enable revocable redactions [33], offering alternative key-rotation mechanisms that complement our proactive secret-sharing and validator-quorum design.

C. Trapdoor Key Sharing and Recovery

To prevent unilateral control, the trapdoor key is distributed using Shamir’s Secret Sharing [34], requiring t -of- n shares for reconstruction. Upon compromise, the contract initiates rekeying and redistributes shares. Proactive secret sharing [35] further mitigates long-term leakage by enabling share refresh without changing the secret, improving resistance to side-channel and slow-leak attacks. Recent work on sanitizable signatures based on lattice-based chameleon hashes offers a complementary path toward post-quantum security in redactable ledgers, which could be integrated into our system in future iterations [36].

D. Privacy-Preserving Logging and Accountability

All redactions are immutably logged on-chain with block and field IDs, pseudonymized user IDs, validator signatures, timestamps, and justification hashes. No plaintext or personal data is stored. Hash commitments enable verifiable audits without revealing sensitive content [37], [38], ensuring non-repudiation and validator accountability.

By combining validation policies, quorum-based consensus, CHF-based mutability, and cryptographic logging, the system ensures strong security and trust guarantees in compliance-centric blockchain applications.

E. Performance Evaluation

To quantify the performance of our redactable blockchain framework, we benchmarked it against both immutable Ethereum baselines and prior redactable systems. While redaction introduces additional logic and state changes, our system achieves better efficiency than earlier prototypes across several key metrics.

Redaction latency averaged 1.82 seconds, broken down into policy validation (0.6s), quorum voting (0.8s), and chameleon hash computation (0.42s). This is significantly faster than the 2.5–3.0s range reported in [39], due to our streamlined middleware and optimized consensus protocol.

The gas cost per redaction was approximately 128,500 units, which is more efficient than the 135k–160k units seen in [15], [39]. Our use of compact metadata in the logging mechanism also helped reduce storage overhead to 1.2%, compared to the 2% average in prior systems.

In terms of consensus performance, the voting delay for a 6-of-10 quorum averaged 0.8s under synchronous conditions and 1.25s under mild network asynchrony, both outperforming the 1.6s delay reported for PoW-based redactions in [16].

These metrics are summarized in Table III for comparison with both immutable baselines and prior redactable models.

Despite modest overhead compared to immutable chains, our design demonstrates improved responsiveness and cost-effectiveness relative to existing redactable solutions, making it viable for deployment in regulated and audit-sensitive domains.

TABLE III
PERFORMANCE COMPARISON: REDACTABLE VS IMMUTABLE

Metric	Immutable	Ours	Prior Work
Redaction Latency	N/A	1.82s	2.5-3.0s [39]
Gas Cost (units)	~45k	128.5k	135k-160k [15]
Storage Overhead	0%	1.2%	1.5-2.0% [15]
Voting Delay	0.3s	0.8s	1.6s [16]

F. Scalability

We evaluated scalability under high-frequency redaction workloads. The system sustained up to 100 redactions per minute with stable throughput; beyond this, latency rose due to congestion in the policy validation contract. This is comparable to Chainspace’s ceiling of 90–95 operations per minute [37]. Future improvements may include parallelizing contract execution or applying sharding to decouple policy checks from validator consensus.

a) Broader Network Considerations: Although tested on a permissioned Ethereum network, broader deployments, such as public or semi-decentralized environments, introduce challenges like validator churn, adversarial behavior, and network latency. Enhancing resilience through proof-of-stake integration, incentive-aligned validator selection, and Byzantine-tolerant quorum mechanisms would improve generalizability.

Section VI further demonstrates how our framework aligns with privacy regulations, including GDPR, by enabling auditable, policy-driven redactions suitable for domains like healthcare, identity, and legal documentation.

VI. COMPLIANCE AND LEGAL MAPPING

Table II summarizes the compliance features of our system with respect to GDPR articles. Through its architectural design and governance model, our redactable blockchain system directly addresses the critical requirements of data protection law, supporting both technical robustness and legal trustworthiness. This positions the framework as a suitable solution for domains where privacy rights and regulatory oversight are essential.

A. GDPR Compliance Alignment

Right to Be Forgotten (Article 17): Our redactable blockchain enables the selective removal of data in accordance with GDPR Article 17, which grants individuals the right to have their personal data erased under specific conditions [1]. Using chameleon hash functions, users can request redaction of blocks containing outdated or unlawful information. The redaction is authorized only after passing policy validation and multi-party approval, ensuring due process. This process eliminates sensitive data while preserving the cryptographic integrity of the chain.

Purpose Limitation (Article 5): The Redaction Policy Engine enforces contextual restrictions on data use. Smart contracts encode whether on-chain data continues to serve its original purpose. If the purpose has expired or changed, redaction can be triggered to comply with the GDPR principle of data minimization and lawful processing.

Accountability and Audit Trails (Articles 5 and 30): The Redaction Log provides a tamper-evident record of all redaction activities, including timestamps, validator signatures, and justification hashes. This supports accountability (Article 5.2) and documentation (Article 30) obligations. Cryptographic signatures and Merkle-based transparency enable regulators to conduct post hoc audits without compromising privacy.

B. Potential Applications

Healthcare Systems: In electronic health record (EHR) management, GDPR compliance is essential. Other frameworks such as HIPAA [2] also impose strict data protection mandates. Our framework allows healthcare providers to redact sensitive patient data upon request or after retention limits are exceeded. For example, a discharged patient’s data can be redacted while preserving cryptographic logs for treatment traceability.

Identity Management Systems: Decentralized Identifiers (DIDs) and Self-Sovereign Identity (SSI) architectures benefit from mechanisms to revoke or delete credentials. Our redactable blockchain supports the controlled deletion of attestations when users withdraw consent or identities become obsolete. This aligns with GDPR’s mandates on consent withdrawal and rectification rights.

Legal Document Chains: In blockchain-based legal archiving systems, our approach allows outdated or voided documents to be removed or corrected without breaking the integrity of the contract chain. Law firms and regulators can use this mechanism to stay aligned with changing privacy and documentation requirements.

VII. CONCLUSION AND FUTURE WORK

This paper presents a redactable blockchain architecture that reconciles regulatory compliance with the foundational principles of decentralization. By combining smart contract-driven policy validation, quorum-based multi-party authorization using BLS signatures, and compatibility with post-quantum chameleon hash functions (PQ-CHF), the system

enables legally governed redactions without compromising cryptographic integrity or governance transparency.

The architecture supports constant-time verification, compact redaction proofs, and throughput exceeding 1,200 requests per second. It meets key GDPR requirements, Articles 5, 17, and 30, while remaining compatible with mainstream platforms like Ethereum.

A modular Redaction Policy Engine (RPE) enforces temporal, legal, and behavioral constraints, and threshold key sharing with immutable audit logs ensures accountability. Unlike prior CHF-based models that rely on static trapdoor holders or lack policy awareness [4], [19], our system offers dynamic, auditable redactions with decentralized enforcement.

While our current implementation uses classical CHF primitives, the architecture is designed to support integration with post-quantum variants such as TMCH [18] and TDPCH [20], enabling time-bound control and traceability. Future work will explore these integrations and extend deployment to public or semi-decentralized networks with incentive-aligned validator selection and Byzantine-resilient consensus. Additionally, recent work on redactable consensus protocols [5] and privacy-preserving smart contracts [23] highlights the growing need for systems that combine legal interpretability with cryptographic rigor. Our framework contributes to this trajectory by bridging policy enforcement with scalable, verifiable redaction logic.

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